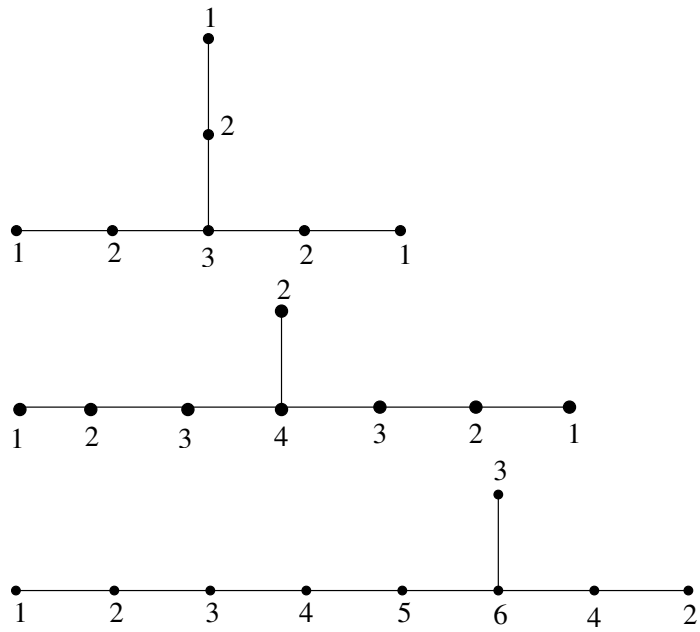

Hint for (c)-(e): What is the meaning of the numbers labeling the vertices of these graphs?

(f) Deduce from (a)—(e) the classification theorem for Dynkin diagrams.

(g) A (simply laced) **affine Dynkin diagram** is a connected graph without self-loops such that the quadratic form defined by A is positive semidefinite but not positive definite. Classify affine Dynkin diagrams. (Show that they are exactly the forbidden diagrams from (c)—(e).)

Problem 6.1.5. Let Q be a quiver with a set of vertices D. We say that Q is of **finite type** if it has finitely many indecomposable representations. Let b_{ij} be the number of directed edges from i to j in Q ($i, j \in D$).

We have the following remarkable theorem, proved by P. Gabriel in the early 1970s.